

Pampanga State University
College of Engineering and Architecture

PILPRO REPORT

Dashboard Visual & Interaction Upgrade

A Project Report

Presented by:
Angelo Abe G. Marasigan

Submitted in partial fulfillment of the requirements
for the BSCPE 4-A Embedded Systems / Robotics Project

2026

Table of Contents

1. Introduction.....	3
2. New Visual & Interaction Features.....	3
2.1 <i>Splash Intro Animation</i>	3
2.2 <i>Toast Notifications</i>	3
2.3 <i>Dark Mode</i>	3
2.4 <i>Next-Dose Progress Ring</i>	3
2.5 <i>Medicine Type Icons</i>	3
2.6 <i>Skeleton Loading States</i>	3
2.7 <i>Empty States</i>	4
2.8 <i>Print-Friendly History View</i>	4
2.9 <i>Onboarding Tour</i>	4
3. Purpose of Each Addition.....	4
4. Current Status and Remaining Work.....	5
5. Conclusion.....	5

1. Introduction

Following the initial version of the PILPRO web dashboard, a set of visual and interaction upgrades was implemented to make the interface feel more polished, easier to understand at a glance, and closer to a production-ready product. This report documents each new feature, why it was added, and what still depends on the real backend and hardware to be fully functional.

2. New Visual & Interaction Features

2.1 Splash Intro Animation

When the dashboard is opened, a full-screen intro now appears first: the word “PILPRO” fades and drops into place one letter at a time against a dark green gradient background, followed by the project tagline. After about 1.5 seconds it fades out and the login screen appears.

2.2 Toast Notifications

Small pop-up messages (“toasts”) now appear briefly at the bottom-right corner of the screen whenever an action is completed — marking a dose as taken, confirming a refill, adding/editing/removing a chamber, exporting history, or clearing notifications — giving the user immediate confirmation that their action succeeded.

2.3 Dark Mode

A toggle in the sidebar switches the entire dashboard between the default light (mint and white) theme and a dark theme, without needing to reload the page. All colors are driven by a single set of theme variables, so every page and component switches consistently.

2.4 Next-Dose Progress Ring

Each chamber card now shows a small circular progress ring next to its dose time, giving a quick visual sense of how soon the next dose is due, and turning red or green once a dose has been missed or taken for the day.

2.5 Medicine Type Icons

Chambers can now be tagged with a medicine type — tablet, capsule, vitamin, or syrup — each shown with its own simple icon on the chamber card and in the schedule form, making it easier to recognize a chamber at a glance instead of relying on text alone.

2.6 Skeleton Loading States

Instead of showing a blank page while data loads, the dashboard briefly displays gray placeholder blocks (“skeletons”) with a shimmer animation in place of the chamber cards,

tables, and notifications, then swaps in the real content shortly after. This keeps the interface feeling responsive even while data is being fetched.

2.7 Empty States

Whenever there are no chambers, no matching history entries, or no notifications, the dashboard now shows a simple icon and short message explaining what to do next, instead of leaving a blank or awkward-looking section.

2.8 Print-Friendly History View

A “Print” button on the History page opens the browser's print dialog with a simplified layout — the sidebar, buttons, and other pages are hidden so only the adherence chart and history table are printed or saved as a PDF directly from the browser.

2.9 Onboarding Tour

After logging in for the first time in a session, a short four-step walkthrough introduces the Dashboard, Schedule, and History/Notifications sections, with “Next” and “Skip” controls, to help a first-time user get oriented quickly.

3. Purpose of Each Addition

These additions were chosen to address common gaps in early-stage prototypes: giving feedback after actions (toasts), handling the in-between moments of loading and empty data gracefully (skeletons and empty states), supporting different lighting/viewing preferences (dark mode), making status easier to read at a glance (progress rings and icons), and supporting real-world use cases like sharing a printed adherence log with a doctor (print view) or introducing a new caregiver to the system (onboarding tour).

Feature	Problem it addresses
Splash animation	Gives the app a distinct, polished identity on open
Toast notifications	Confirms that an action (mark taken, refill, save) succeeded
Dark mode	Comfort in low-light use and personal preference
Progress ring	Faster read of “how soon is the next dose” than text alone
Medicine icons	Quicker visual recognition of chamber contents
Skeleton loading	Avoids a blank/broken-looking screen while data loads
Empty states	Explains clearly what to do when there is no data yet
Print view	Lets a user share or archive an adherence log easily
Onboarding tour	Orients a first-time user without external instructions

4. Current Status and Remaining Work

All of the features above are implemented and working on sample data within the browser, consistent with the dashboard's current stage as a front-end-only prototype. As with the rest of the dashboard, they do not yet reflect real device or account data. Remaining work specific to this visual layer includes:

- Confirming that the dark mode color palette meets accessibility contrast guidelines once real content and branding are finalized.
- Deciding whether the onboarding tour should only appear once per real user account (requiring backend/account support) instead of once per session.
- Testing the print view against different browsers and paper sizes once real history data is available.
- Expanding the medicine type icon set if additional medication forms (e.g., inhalers, drops) are needed.

5. Conclusion

This visual and interaction upgrade brings the PILPRO dashboard noticeably closer to a polished, user-ready interface while remaining fully front-end and demo-friendly. Combined with the data layer's existing readiness for backend integration, the dashboard is now positioned to move into real device and account integration with its user experience largely already in place.